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FMRAG: retrieval-augmented multimodal large language models for fisheries intelligence

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Introduction: Multimodal large language models (MLLMs) have exhibited significant potential for fisheries analysis. However, their inherent hallucination issues in species identification, ecological activity interpretation, environmental assessment, and biomass estimation severely restrict their reliability and practical application in real-world fisheries management.

Methods: This study proposes FMRAG, a fisheries-oriented multimodal retrieval-augmented generation framework to enhance factual grounding and domain adaptability. The framework retrieves visually similar fishery images and corresponding textual records within a unified vision-language embedding space, and integrates retrieved multimodal evidence with query inputs via a cross-modal fusion mechanism to exploit fine-grained visual features and domain-specific textual knowledge. Three fisheries-tailored fine-tuning tasks are further introduced: image-text association learning, visual concentration learning, and retrieval-augmented reasoning learning, to strengthen multi-image reasoning and multimodal alignment.

Results: Experimental results on species identification and biomass estimation demonstrate that FMRAG consistently outperforms baseline MLLMs and text-only RAG methods, effectively reducing hallucinations and improving predictive accuracy. The proposed framework also shows superior performance in rare-species recognition, temporal stability, and confidence calibration, and can be successfully transferred to models originally trained on single-image inputs.

Discussion: FMRAG provides an effective and practical solution for constructing trustworthy multimodal intelligence systems, supporting reliable and robust applications in fisheries monitoring and management.

KEYWORDS
fisheries resource management, hallucination mitigation, multimodal large language models, retrieval-augmented generation, sustainable fishing

1 Introduction

In recent years, multimodal large language models (MLLMs) have attracted increasing attention for their ability to integrate heterogeneous information sources and support complex decision-making in scientific and industrial domains, including ecology and fisheries (Liu et al., 2024; Zhang et al., 2023). By jointly modeling visual and textual information, these models offer new opportunities for fishery resource monitoring, biomass assessment, habitat interpretation, and management-oriented reporting (Guu et al., 2020).

Such capabilities are particularly valuable as fisheries worldwide face mounting pressure from climate change, ecosystem degradation, and intensified human activity, all of which demand more timely and data-driven management strategies (Johnson et al., 2020). Although large language models have shown potential in ecological analysis and policy support (Pons et al., 2020), their reliability remains a critical concern in high-stakes fishery applications, where factual errors may lead to misleading stock assessments, inappropriate management recommendations, or even ecological harm (Tanaka et al., 2023).

Despite this promise, current MLLMs still suffer from substantial limitations when applied to fisheries scenarios. Fisheries management depends on the accurate interpretation of multimodal evidence (Chen et al., 2021), including underwater images, video sequences, textual records, and environmental sensor measurements (Liu et al., 2023). Existing studies have explored MLLMs for tasks such as fish species identification, biomass estimation, and habitat assessment (Qu et al., 2024; Chust et al., 2014; Klemas, 2013), yet these models frequently generate hallucinated outputs, including species misidentification, implausible biomass estimates, environmentally inconsistent conclusions, and erroneous regulatory suggestions (Smith et al., 2022; Huang et al., 2024). Retrieval-Augmented Generation (RAG) has been widely adopted to improve factual grounding by introducing external knowledge during inference (Zhang et al., 2023; Siriwardhana et al., 2022). However, most existing RAG frameworks in domain applications remain text-centric (Chollett et al., 2022), which is inadequate for fisheries because key evidence is often inherently visual or jointly determined by visual and environmental cues (Pauly and Zeller, 2016; Raffel et al., 2020). Moreover, prior multimodal retrieval approaches often rely on text labels or captions as proxies for image content (Watson and Tidd, 2018; Costello et al., 2016), which can be insufficient for distinguishing visually similar fish species or interpreting subtle ecological patterns. In parallel, MLLMs are still weak in multi-image reasoning (Ribeiro et al., 2016), especially when they are pre-trained mainly on single-image inputs, making it difficult for them to fully exploit retrieved multimodal evidence (Sumaila et al., 2021; Hilborn et al., 2020). Existing hallucination mitigation methods also focus predominantly on text-only verification or prompt engineering (Yue et al., 2013; Malkov and Yashunin, 2016; Brown et al., 2020), and therefore do not adequately address the cross-modal consistency requirements of fisheries analysis.

To address these limitations, this study proposes FMRAG, a Fishery Multimodal Retrieval-Augmented Generation framework that explicitly integrates both visual and textual evidence for fisheries-oriented reasoning. Instead of retrieving images through textual surrogates alone, FMRAG retrieves image–text pairs directly in a shared vision–language embedding space using the query image as the primary retrieval key. The retrieved multimodal evidence is then fused with the query input to support grounded inference on species recognition, biomass estimation, environmental interpretation, and related fishery tasks. To further enhance the ability of MLLMs to use retrieved multimodal context, we design three fisheries-tailored fine-tuning tasks, namely Mloc, Mcon, and Mfmrag, which strengthen image–text correspondence learning, visual concentration under noisy conditions, and retrieval-augmented multi-image reasoning. In addition, we introduce a multimodal hallucination detection module that verifies generated claims through consistency checks across retrieved images, textual records, and sensor-related evidence. In this way, the proposed framework is intended not only to improve prediction accuracy, but also to enhance the trustworthiness and practical usability of MLLMs in fisheries management scenarios. Figure 1 presents representative

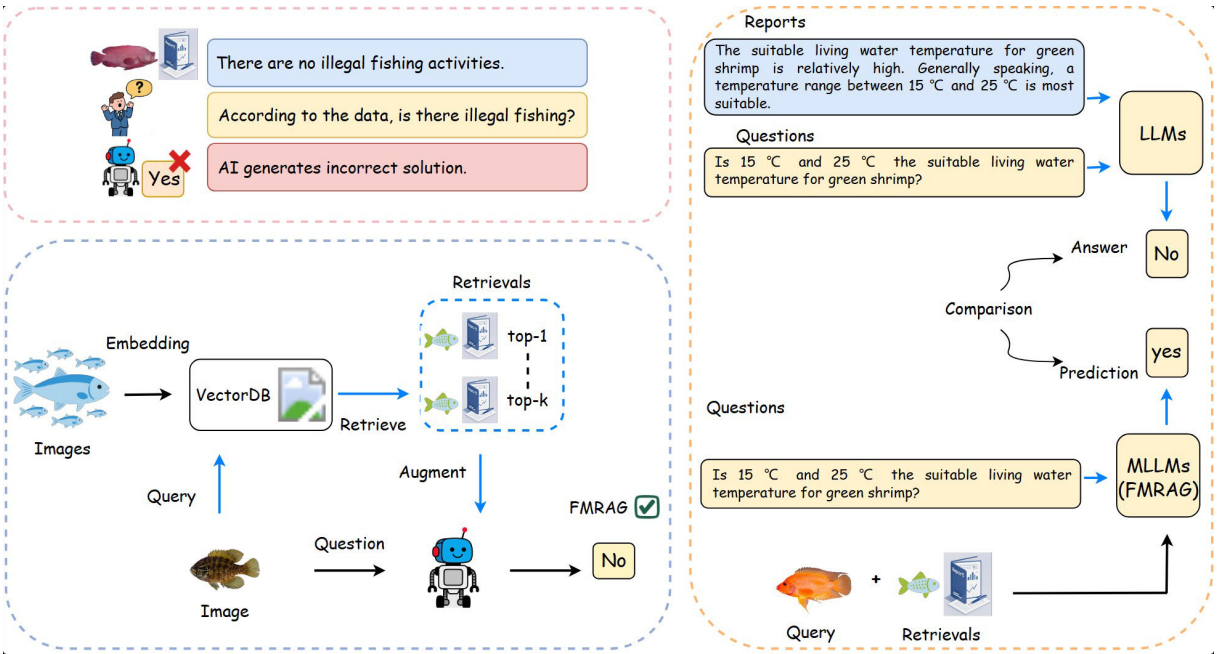


FIGURE 1
Examples of hallucinated outputs from a baseline MLLM in fisheries tasks (Left UP). FMRAG improves response accuracy by retrieving the Top-1 most relevant image–report pair from a vector database and using it as context for generation. Propose fish species-based questions to FMRAG-based MLLM and compare the predictions with the answers in LLM's interpretation of the reference report.

failure cases of baseline fishery MLLMs and the corresponding correction mechanism enabled by FMRAG.

The main contributions of this work are summarized as follows.

1. We propose a fisheries-oriented multimodal retrieval-augmented generation framework, FMRAG, which retrieves visually similar fishery images together with associated textual records and integrates them into a unified reasoning pipeline.
2. We introduce three domain-specific fine-tuning tasks, Mloc, Mcon, and Mfmrag, to improve the ability of MLLMs to align images with reports, focus on task-relevant visual content, and perform retrieval-enhanced multi-image reasoning.
3. We construct and evaluate the framework on a benchmark of 5,000 annotated multimodal fisheries samples and conduct extensive experiments covering overall identification performance.

The remainder of this paper is organized as follows. Section 2 reviews related work on multimodal retrieval-augmented generation, multi-image reasoning, and hallucination mitigation. Section 3 presents the proposed FMRAG framework, including multimodal retrieval, cross-modal integration, fisheries-specific fine-tuning tasks, and the hallucination detection module. Section 4 reports the experimental settings and results, including ablation studies, robustness analyses, rare-species evaluation, biomass estimation analysis, and practical deployment considerations. Finally, Section 5 concludes the paper and discusses future directions for trustworthy multimodal intelligence in fisheries management.

2 Related work

2.1 Multimodal retrieval-augmented generation for domain-specific reasoning

Previous studies on multimodal retrieval-augmented generation have mainly followed two representative lines, both of which reveal limitations in domain-specific scenarios such as fisheries science. The first line retrieves visual content through associated textual metadata. For example (Watson and Tidd, 2018), and (Costello et al., 2016) showed that retrieving images based on text labels or captions can improve performance in general-domain visual question answering. However, this strategy relies on the strong assumption that textual descriptions fully capture the semantic content of the image. Such an assumption is often invalid in fisheries applications, where subtle visual differences, such as body shape, fin structure, or color pattern, may be critical for distinguishing phenotypically similar species but may be only partially described or entirely omitted in text. To address this issue, our framework retrieves image-text pairs directly within a shared vision-language embedding space and uses the query image itself as the retrieval key, thereby reducing dependence on incomplete text-based proxies.

The second line extends retrieval-augmented generation to domain applications while still remaining predominantly text-centric. Prior studies such as (Vaswani et al., 2017) and (Wang et al., 2024; Touvron et al., 2021) developed retrieval pipelines for ecological or visual understanding tasks, but they generally assumed that retrieved text alone provides sufficient contextual grounding. This assumption is inadequate for fisheries reasoning, where correct interpretation often depends on the joint presence of visual, textual, and environmental evidence. Our results challenge this text-only paradigm through systematic ablation analysis. On the FishVista dataset, text-only retrieval achieves an F1 score of 0.712, whereas full multimodal retrieval improves the score to 0.723 ($p < 0.05$). Although the absolute gain appears moderate, detailed failure analysis shows that text-only retrieval consistently underperforms when species descriptions are ambiguous or when decisive evidence resides in visual features that text cannot adequately encode. These observations indicate that effective fisheries reasoning requires multimodal retrieval rather than the simple extension of text-based RAG.

2.2 Multi-image reasoning and hallucination mitigation in multimodal large language models

Another closely related research direction concerns the reasoning limitations of multimodal large language models when processing multiple images. Previous studies, including (Sumaila et al., 2021) and (Hilborn et al., 2020), reported that MLLMs often struggle to integrate information across multiple visual inputs and suggested that this weakness may arise from architectural constraints. In contrast, the present study argues that at least part of this limitation stems from insufficient task-specific training rather than from the architecture alone. Through the proposed fine-tuning tasks, namely Mloc (image-text association), Mcon (image concentration), and Mfmrag (multimodal retrieval-augmented generation), even models originally pre-trained on single-image inputs can acquire stronger multi-image reasoning capabilities. This perspective shifts the emphasis from architectural redesign to targeted adaptation, which is particularly important in fisheries applications where retrieved visual evidence must be jointly interpreted with reports and environmental context.

Closely related to this issue is the growing literature on hallucination mitigation in large language models. Existing studies such as (Yue et al., 2013; Malkov and Yashunin, 2016) and (Brown et al., 2020) provide broad discussions of hallucination reduction strategies, most of which focus on prompt engineering, output filtering, or text-only verification. While these approaches are useful in general language settings, they are insufficient for fisheries analysis, where the validity of a prediction often depends on consistency across heterogeneous modalities. In response to this limitation, the hallucination detection module proposed in this study adopts a fundamentally different strategy by performing multimodal consistency checking over retrieved images, textual records, and sensor-related evidence. Rather than relying solely on textual fact verification, our method exploits the complementary

and mutually validating nature of heterogeneous data sources. This design is especially suitable for fisheries science, where accurate reasoning often depends on whether visual observations, environmental conditions, and report content support the same conclusion.

3 Methodology

The proposed FMRAG framework is designed to improve the reliability of Fishery-MLLMs by integrating multimodal retrieval, cross-modal evidence fusion, fisheries-specific fine-tuning, and post-generation hallucination verification into a unified pipeline. As shown in Figure 2, the framework first encodes fisheries images, textual records, and sensor-related information into a shared representation space, and then retrieves the most relevant multimodal references for a given query image. These retrieved references are incorporated into the model input to provide external evidence for downstream reasoning. On this basis, the framework further enhances the model through task-specific fine-tuning and finally applies a hallucination detection module to verify whether the generated response is consistent with the retrieved evidence. Through this design, FMRAG addresses three core challenges in fisheries analysis, namely insufficient multimodal grounding, weak multi-image reasoning, and the generation of hallucinated outputs.

3.1 Retrieval of MLLM

The retrieval component is designed to fetch fisheries-specific images and records that are semantically related to the input query by jointly exploiting visual and textual information. For a given fish image F_img, we first extract its image embedding FE_img∈R^d using an embedding model, where d denotes the embedding dimensionality. The extracted embedding is then stored in a memory store MS constructed with Chroma, which serves as a vector database for efficient semantic indexing and search (Lundberg and Lee, 2017). To retrieve relevant image-text pairs, we employ the Hierarchical Navigable Small World (HNSW) algorithm (Howard et al., 2019), which compares the embedding of the query image with all stored entries and returns the top-k nearest neighbors in the vector space. The Top-1 result is regarded as the closest match and is used as the primary context for response generation, while the remaining retrieved items provide supplementary evidence.

The retrieved outputs contain both visual references and their associated textual records, enabling the model to access fisheries-

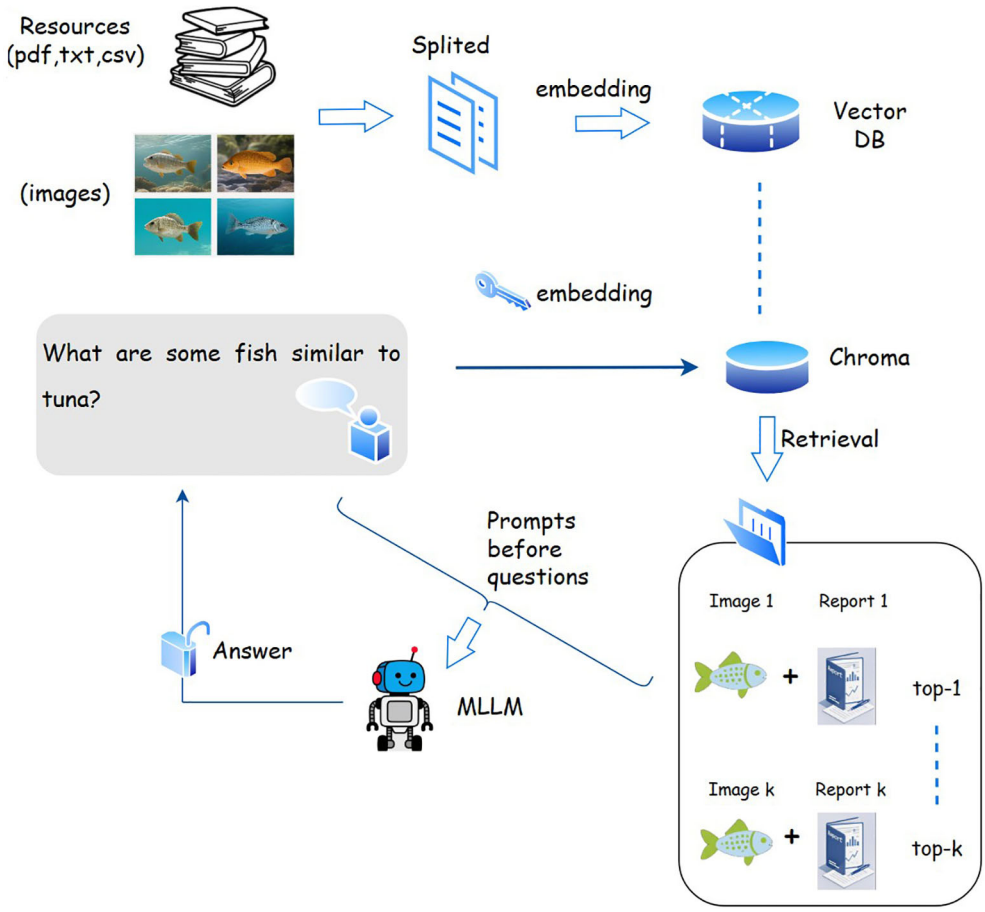


FIGURE 2
Overview of the proposed FMRAG framework for fisheries-oriented multimodal reasoning. The framework integrates multimodal retrieval, cross-modal fusion, fisheries-specific fine-tuning, and hallucination detection.

specific contextual evidence rather than relying solely on internal parametric knowledge. We adopt the HNSW implementation in Chroma because it offers strong performance under high-recall retrieval settings and scales effectively to larger databases. Compared with IVF style indexing, which requires cluster training and may suffer from quantization error, HNSW builds a proximity graph that supports more accurate nearest-neighbor search without the need for repeated retraining. For the current dataset size of approximately 5000 entries, and for anticipated future deployment scenarios involving tens or hundreds of thousands of entries, HNSW provides a favorable balance between retrieval speed and search accuracy.

For image encoding, we use a CLIP-based model, specifically openai/clip-vit-large-patch14, to extract visual features. For textual records such as reports and metadata, we use the corresponding CLIP text encoder so that images and text are projected into a unified embedding space in which cosine similarity is semantically meaningful. Numerical sensor data are first normalized and then projected into the same space through a learned linear layer before being concatenated with the visual representation. Similarity between the query image embedding and the stored entries is computed using cosine similarity, which is also the default metric of the HNSW index. In all primary experiments, we set k to 5. This choice reflects a balance between retrieval completeness and the input token limitations of the MLLM, while also matching the empirical observation that retrieval performance begins to plateau beyond this point.

3.2 Data integration

After retrieval, the next step is to integrate heterogeneous evidence into a unified multimodal representation for downstream reasoning. In the proposed framework, sensor data such as water temperature and pH are first normalized and encoded into a numerical feature vector. This vector is concatenated with the visual embedding extracted from the query image, forming a joint query-side representation. The resulting multimodal representation is then passed through a cross-modal attention layer, which dynamically assigns modality-specific importance weights according to the contextual demands of the task. For example, species identification relies more heavily on visual morphology, whereas environmental anomaly detection may depend more strongly on sensor measurements. In this way, the framework supports adaptive evidence fusion rather than treating all modalities as equally informative.

Formally, the concatenated representation of the sensor vector s and the query image embedding v_q serves as the Query, while the Keys and Values are derived from the embeddings of the retrieved images {v_1…v_k} and their associated text reports {t_1…t_k}. Through this design, the model can jointly exploit the current query image, environmental measurements, and retrieved visual-textual references within a unified attention-based fusion process. During inference, the query fish image Fq is first encoded into an embedding, after which the system retrieves the top-k images (I1,…, Ik) and their associated fishery reports (FR1,…, FRk) from the memory store MS. These references are then inserted before the final question according to the prompt organization shown in Figure 3, allowing the Fisheries-MLLM to generate answers conditioned on both the query input and external multimodal context. By integrating visual and textual evidence in this way, the model gains access to complementary cues that improve factual grounding and reduce the likelihood of hallucinated responses (Tan and Le, 2019; Cinner et al., 2012).

3.3 Enhancing Fishery-Mllms by fine-tuning

Although retrieval provides external evidence, some Fishery-MLLMs may still be insufficiently trained to distinguish and reason over multiple fish images within a single context (Wilkinson et al., 2016). To address this limitation, we further fine-tune the model on fisheries-specific tasks such as fish species identification and biomass assessment, thereby adapting it to the multimodal reasoning requirements of this domain (Kim et al., 2024; Moon et al., 2024). As illustrated in Figure 4, we design three fine-tuning tasks to strengthen image-text association, attention control, and retrieval-aware inference within the FMRAG framework.

We begin with a base dataset defined as S = (img_i, P_i, A_i), where img_i denotes the i-th fish image, P_i denotes the corresponding prompt, A_i denotes the answer, and i ranges from 1 to N. Based on this dataset and the memory store MS, we construct three task-specific training sets. The first task, Mloc, is designed to improve image-text correspondence learning. For this task, we sample K images ($1 \leq K \leq 5$) from different fisheries scenarios together with their corresponding reports Rij. Each training instance consists of an image set (img_i1,…, img_iK), a textual query

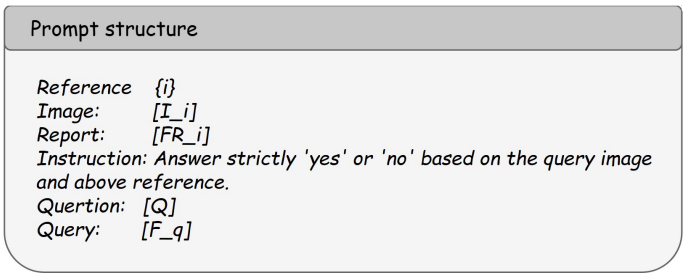


FIGURE 3
Image retrieval prompt structure.

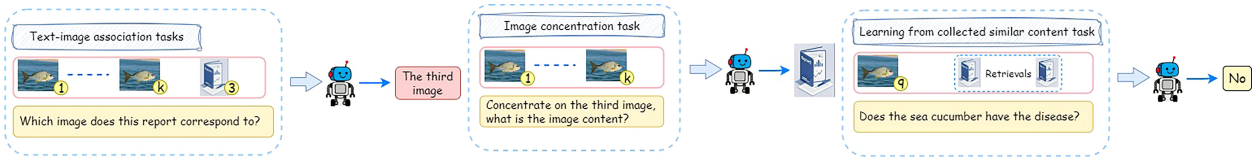


FIGURE 4
Empower the fishery MLLM FMRAG capability through fine-tuning.

P'_ij = "Identify which image (1-K) matches: [Rij]", and a target label A'_ij indicating the correct image position. This task encourages the model to learn fine-grained visual-textual alignment, positional reasoning among multiple candidate images, and the interpretation of fisheries-specific semantics such as species and habitat descriptions. Because K is varied from 1 to 5, the task gradually increases in difficulty while remaining ecologically realistic.

The second task, Mcon, is designed to improve visual attention under multi-image conditions. In this task, the model is required to generate or interpret content for a designated image within a K-image set using prompts of the form P"_ij = "Focus on the j-th image: [P_ij]". This design teaches the model to suppress interference from irrelevant images and maintain single-image analytical precision even when multiple images are simultaneously present. The third task, Mfmrag, directly simulates the actual inference process of FMRAG. A query image img_q is paired with its top-K retrieved references {(img_q1, A_q1),…, (img_qK, A_qK)}, and the model is trained using prompts of the form P"'_q = "Based on the query image and references: [P_q]". This task strengthens cross-image reasoning and retrieval-aware decision-making, which is especially important for fisheries diagnosis and question answering tasks such as determining whether a given image contains a particular species. Taken together, Mloc strengthens image-text association, Mcon enhances focused visual reasoning, and Mfmrag improves retrieval-augmented inference, allowing Fishery-MLLMs to better exploit the multimodal evidence provided by FMRAG.

3.4 Hallucination detection module

To further improve output reliability, we introduce a hallucination detection module as a post-processing stage. This module takes as input the generated response R of the MLLM together with the retrieved multimodal evidence set E, which consists of the top-k retrieved images and reports. Its purpose is to determine whether the generated content is supported by the retrieved evidence or whether it contains hallucinated claims that should be corrected before final output.

The first step is fact extraction. A lightweight fine-tuned named entity recognition model identifies key factual claims from R, such as fish species names, environmental values, and descriptions of activities or conditions. The second step is evidence verification. For each extracted claim c, the module queries the evidence set E. If the claim is textual in nature, such as a statement about water temperature, the module performs semantic matching against the retrieved reports. If the claim is visual in nature, such as a species classification, the module relies on a specialized fish species classifier applied to the retrieved images. The third step is consistency checking. A claim is marked as hallucinated if it is directly contradicted by the retrieved evidence, or if no supporting evidence can be found for a domain-specific factual statement that should in principle be verifiable, such as a catch quota or habitat condition. The hallucination detector was validated on a held-out set of 200 annotated examples and achieved an F1 score of 0.89. When hallucination is detected, the system triggers a response revision stage, instructing the MLLM to regenerate the answer while focusing strictly on the retrieved multimodal evidence. Through this mechanism, the framework provides an additional layer of factual control and further improves the trustworthiness of fisheries-oriented multimodal reasoning. MLLM to regenerate the answer with a specific instruction to focus only on the retrieved evidence."

4 Experiment

4.1 Datasets and evaluation metrics

To evaluate the effectiveness of FMRAG, we constructed a fisheries-oriented multimodal dataset containing 5000 annotated underwater images, each paired with textual descriptions and environmental metadata such as water temperature and pH. The dataset was collected from two publicly available sources, namely FishVista and Fish4Knowledge. FishVista images were obtained from the IUCN Photo Library and research institutions under academic-use licenses, while Fish4Knowledge data were collected from underwater cameras in Taiwan under a research-use license. All images were used in accordance with their licensing terms. The final dataset covers 580 species and exhibits a clear long-tail distribution, as summarized in Table 1. The top 20 species account for 62.4 percent of all images, whereas 560 rare species share only 37.6 percent of the data, highlighting the practical challenge of long-tail recognition in fisheries applications.

To construct the multimodal question-answer evaluation set, we followed a five-step procedure. First, fish species entities were extracted from fisheries reports using a fine-tuned recognition model. Second, Qwen2.5-VL-7B-Instruct was used to generate standardized yes or no questions for each identified entity, such as "Does the image show [species] based on [Report]?" Third,

TABLE 1 Distribution of fish species in the evaluation dataset.

Species category	Number of species	Total images	Percentage of dataset (%)	Average images per species
Top-10 Most Frequent	10	2,471	49.4%	247.1
Top-11 to Top-20	10	650	13.0%	65.0
Top-20 Total	20	3,121	62.4%	156.1
Rare Species	560	1,879	37.6%	3.4
Overall	580	5,000	100%	8.6

irrelevant categories such as equipment or non-biological entities were removed. Fourth, closely related taxonomic labels were consolidated to avoid over-fragmentation. Fifth, species terms were normalized to reflect realistic occurrence frequency. Following this procedure, we obtained 650 multimodal question-answer pairs from FishVista and 1200 from Fish4Knowledge, containing 280 and 550 unique species, respectively. Species identification performance was evaluated using Precision, Recall, and F1 score. In addition, the biomass estimation task was assessed using Mean Absolute Percentage Error (MAPE), Interval Coverage Probability (ICP), and Temporal Consistency Error (TCE). Representative examples of the dataset and fine-tuning samples are shown in Figure 5.

For model training, the data were further organized into a structured dialogue format so that image paths and paired descriptions could be directly used by Qwen2.5-VL-7B. Specifically, each sample was converted into a JSON instance containing the image path and its corresponding response text, allowing seamless integration into the multimodal instruction-tuning pipeline.

4.2 Experimental setups

All experiments were conducted on Qwen2.5-VL-7B, which served as the base multimodal large language model for evaluating both retrieval augmentation and fisheries-specific fine-tuning. To adapt the model efficiently, we employed the Low-Rank Adaptation (LoRA) strategy on Qwen2.5-VL-7B-Instruct. The fine-tuning hyperparameters were set as follows: batch size 8 per GPU, 3 training epochs for all primary tasks, LoRA rank r = 16, LoRA alpha α = 32, learning rate 1e-4 for LoRA parameters, and weight decay 0.01 for all non-bias and non-LayerNorm parameters. This setup was used consistently across all main experiments.

The experiments were implemented in Python 3.10 and executed on two NVIDIA RTX 4090 GPUs with 24 GB memory each. Chroma was used as the vector database for multimodal retrieval, while LangChain and Ragas were employed for pipeline construction and evaluation. During inference, the end-to-end processing of a single query, including image embedding, Top-5 retrieval, and answer generation, required approximately 1.8 seconds under the dual GPU configuration. This latency is acceptable for offline decision support and benchmarking, although further optimization would be required for real-time deployment scenarios.

4.3 Experimental results

We first evaluated overall species identification performance under four settings, namely baseline MLLM, MLLM+RAG with text-only retrieval, FMRAG with multimodal retrieval, and fine-tuned FMRAG. As shown in Table 2, the baseline MLLM achieved relatively high precision but very low recall across all three datasets, indicating that the model was conservative but unable to recover a sufficient number of true positives. Text-only RAG improved recall and overall F1 score, confirming that external textual grounding is beneficial, but its performance remained clearly below that of multimodal retrieval. FMRAG substantially improved recall while maintaining competitive precision, leading to much stronger F1 scores on FishVista, Fish4Knowledge, and WildFish. Fine-tuned



FIGURE 5
A random image selection from the datasets (Left). Typical fine-tuning dataset image (Right).

TABLE 2 Overall fish species identification performance for comparison of four methods across three fisheries datasets.

Method	FishVista			Fish4Knowledge			WildFish		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
MLLM	0.911	0.186	0.361	0.952	0.290	0.412	0.951	0.280	0.442
MLLM+RAG	0.860	0.377	0.515	0.942	0.536	0.644	0.941	0.524	0.544
FMRAG	0.752	0.903	0.711	0.950	0.915	0.882	0.948	0.914	0.862
FMRAG (fine-tuned)	0.781	0.921	0.743	0.951	0.959	0.910	0.950	0.940	0.890

FMRAG further improved performance on all datasets and achieved the best results overall, demonstrating the importance of fisheries-specific fine-tuning for effective use of retrieved multimodal evidence.

To ensure the reliability and consistency of the multimodal annotations, a structured annotation protocol was established. All samples were first pre-processed to align underwater images, textual descriptions, and environmental metadata. Species labels and related semantic information were then annotated following standardized fisheries taxonomy guidelines. The annotation process was conducted by three annotators with academic backgrounds in marine science and computer vision, each having prior experience in fish species identification and ecological data analysis. During the initial phase, a subset of the dataset was jointly reviewed to calibrate annotation criteria and resolve ambiguities in species naming and ecological descriptions. Each sample was subsequently labeled independently by two annotators, and disagreements were resolved through discussion or adjudication by a senior reviewer with domain expertise. To further ensure annotation quality, periodic consistency checks were performed throughout the annotation process, and ambiguous cases were documented and reviewed collectively. This protocol helped maintain high annotation reliability and ensured that the resulting dataset provided accurate multimodal supervision for training and evaluation.

We then examined the effectiveness of multimodal retrieval itself. Table 3 shows that retrieval recall improves from Top-1 to Top-3 across all fisheries tasks, while the relative importance of image and text varies by scenario. Species identification depends predominantly on visual information, whereas disease inquiry benefits more from textual support. This confirms that multimodal retrieval provides complementary evidence rather than redundant information. Figure 6 further shows that retrieval recall increases with k and gradually saturates, supporting the use of Top-5 retrieval as a practical balance between performance and efficiency.

TABLE 3 Multimodal retrieval recall verification table.

Type	Top-1 recall (%)	Top-3 recall (%)	Modal
Species Identification	74.34	76.82	Image 80%, Text 20%
Environmental Prediction	65.48	72.91	Image 60%, Text 40%
Activity Analysis	64.48	73.23	Image 60%, Text 40%
Disease Inquiry	66.48	73.91	Image 40%, Text 60%

To assess practical usefulness, we further evaluated FMRAG in representative fisheries scenarios, including species recognition, environmental prediction, regulatory inquiry, and abnormal detection. As shown in Table 4, the framework achieved the best precision in species identification and environmental prediction, while maintaining relatively low hallucination rates across tasks. These results indicate that FMRAG can support a variety of fisheries applications beyond species classification alone.

To analyze the effect of species frequency, we divided the dataset into high-frequency and low-frequency species groups and evaluated the performance of different models under long-tail conditions. Figure 7 and Figure 8 show that FMRAG consistently achieves the highest F1 scores for both frequent and rare species, while the gap becomes especially large for low-frequency categories. This indicates that multimodal retrieval is particularly valuable for handling rare and difficult species, where parametric knowledge alone is insufficient.

We also investigated whether the proposed fine-tuning strategy can transfer FMRAG capability to models originally trained on single-image inputs. Starting from a single-image version of Qwen2.5-VL-7B, we fine-tuned the model with different task combinations. As shown in Table 5, the baseline single-image model already achieved reasonable precision, but its recall was much lower than that of the task-adapted variants. All three fisheries-specific fine-tuning tasks improved F1 score, and the combined setting of Mloc, Mcon, and Mfmrag achieved the best result. This confirms that the proposed training strategy effectively equips single-image-pretrained models with retrieval-augmented multi-image reasoning ability.

To verify statistical robustness, all major experiments were repeated with five random seeds. Table 6 shows that the standard deviations of all reported metrics remained below 0.02, indicating stable and reproducible performance. The improvements of fine-tuned FMRAG over baseline methods remained statistically significant across repeated trials.

Finally, we evaluated the end-to-end hallucination detection module. Table 7 shows that FMRAG achieved the highest detection accuracy and mitigation success rate while maintaining the lowest false positive rate among all compared methods. Table 8 further shows that the gain is consistent across multiple hallucination types, including species misidentification, environmental inconsistency,

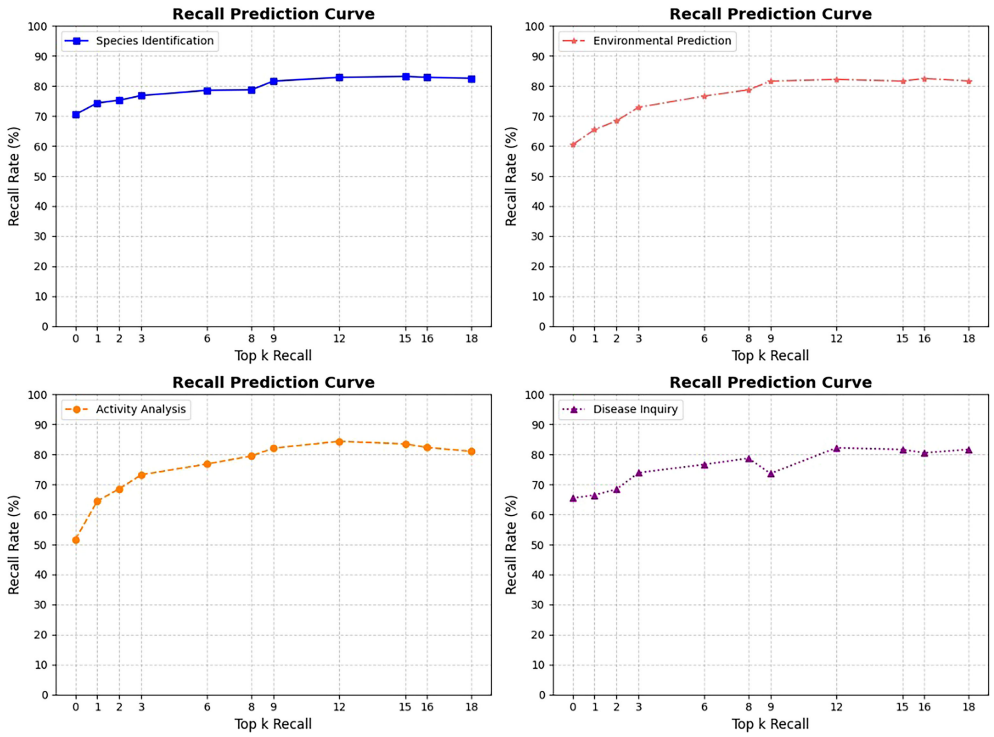


FIGURE 6
Recall prediction curve based on top k recall.

regulatory error, and biomass overestimation. These results confirm that multimodal consistency checking is more reliable than text-only hallucination verification in fisheries scenarios.

FMRAG demonstrates superior performance across all hallucination types, with the largest gains in species misidentification (92.1% vs. 74.2% for SelfCheckGPT) and environmental inconsistency (90.4% vs. 76.8%). This confirms that multimodal consistency checking is particularly effective for domain-specific hallucinations that require cross-referencing visual, textual, and sensor evidence.

4.4 Ablation experiments

To understand the contribution of different modalities, we conducted ablation experiments by progressively removing components from the full FMRAG framework. As shown in Table 9, the full multimodal configuration achieved the highest performance in both species identification and environmental analysis while also producing the lowest hallucination rate. Removing sensor data reduced overall performance, and relying on text alone caused a substantial drop in both accuracy and factual reliability. These results demonstrate that visual, textual, and environmental inputs contribute complementary evidence and that their joint use is crucial for fisheries reasoning.

TABLE 4 Domain scenario performance testing table.

Type	Precision (%)	Hallucination (%)	Modal
Species Identification	94.2	5.1	Image+Text
Environmental Prediction	88.7	8.3	Text+Sensor data
Regulatory Inquiry	79.4	12.6	Sensor data+ Image
Abnormal Detection	69.3	10.5	Sensor data+ Image

We further analyzed the effect of retrieval modality itself. As shown in Table 10, image-only retrieval already improved F1 score substantially over the baseline model, indicating that visual nearest-neighbor retrieval is effective for fisheries tasks. Text-only retrieval performed slightly better, likely because taxonomic names and descriptions directly support species recognition. However, the best performance was achieved when both image and text retrieval were used together, confirming that multimodal retrieval provides the strongest grounding.

To evaluate the contribution of the proposed fine-tuning tasks, we compared different combinations of Mloc, Mcon, and Mfmrag. As shown in Table 11, the Mfmrag task alone already improved performance over the unfine-tuned setting, indicating that retrieval-aware learning is beneficial. Adding Mcon led to further gains, while adding Mloc produced a slightly larger improvement. The best results were obtained when all three tasks were used together, showing that image-text alignment, visual focus, and retrieval-enhanced reasoning are complementary.

4.5 Computational efficiency analysis

To assess the practical cost of multimodal retrieval, we measured the end-to-end inference latency of different configurations

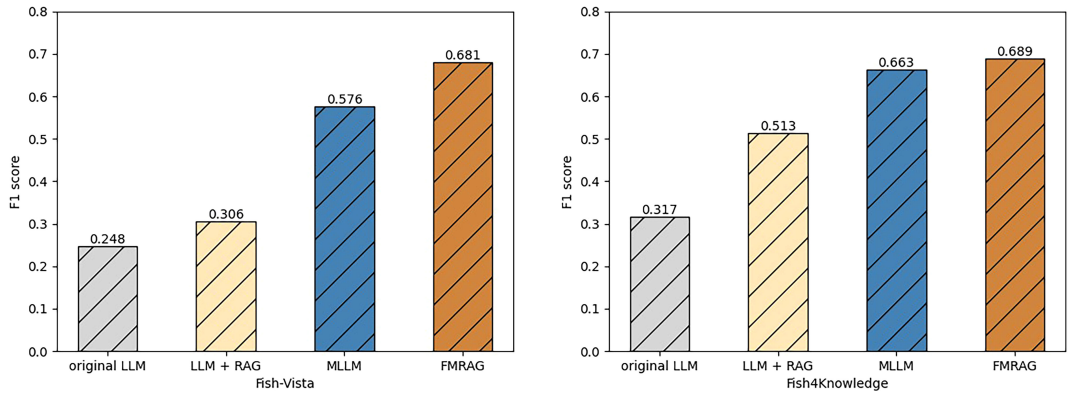


FIGURE 7
High-frequency fish species identification performance.

on the dual RTX 4090 setup. As shown in Table 12, the baseline MLLM required 850 ms per query, while FMRAG with k = 5 required 1100 ms. Most of the extra cost came from image embedding and vector retrieval rather than text generation. Reducing k from 5 to 3 or 1 decreased latency while preserving much of the performance gain, indicating that retrieval depth can be adjusted depending on the application scenario.

To place FMRAG within the broader MLLM landscape, we also compared it against several strong open-source multimodal models, including LLaVA-NeXT-8B, CogVLM2-19B, InternVL2-8B, and Qwen2.5-VL-7B. As shown in Table 13, all models benefited substantially from FMRAG, and all achieved further gains after fisheries-specific fine-tuning. This demonstrates that the proposed framework is model-agnostic and transferable across different architectures. Although larger models achieved slightly better peak performance, the results indicate that retrieval design and task-specific adaptation are more important than raw model scale for fisheries applications.

We then examined retrieval depth more systematically. Figure 9 shows that F1 score increases rapidly as k grows from 1 to 5 and then begins to saturate. Figure 10 shows a similar trend from the perspective of retrieval recall. Together, these results justify the use of k = 5 as a practical default setting.

We further analyzed the types of errors reduced on rare species. Figure 11 presents confusion matrix comparison for rare species (top-10 rare species). FMRAG demonstrates a substantial reduction in three critical error types for rare species. Errors arising from confusion with visually similar common species fell by 67.3% (from 342 to 112). False negatives, or missed detections, decreased by 58.9% (from 418 to 172). Furthermore, instances of hallucinated rare species in images of common species were reduced by 71.4% (from 98 to 28).

We also focused on rare species identification, which is particularly important for biodiversity monitoring. As shown in Table 14, FMRAG and fine-tuned FMRAG dramatically improved F1 score for rare species compared with both the baseline MLLM and text-only RAG. The improvement for rare categories was much larger than that for high-frequency species, demonstrating that multimodal retrieval is especially valuable for long-tail data. Figure 12 further confirms that FMRAG consistently outperforms the baseline MLLM across the 20 rarest species.

We evaluated biomass estimation performance. As shown in Table 15, FMRAG substantially improved F1 score, reduced MAPE,

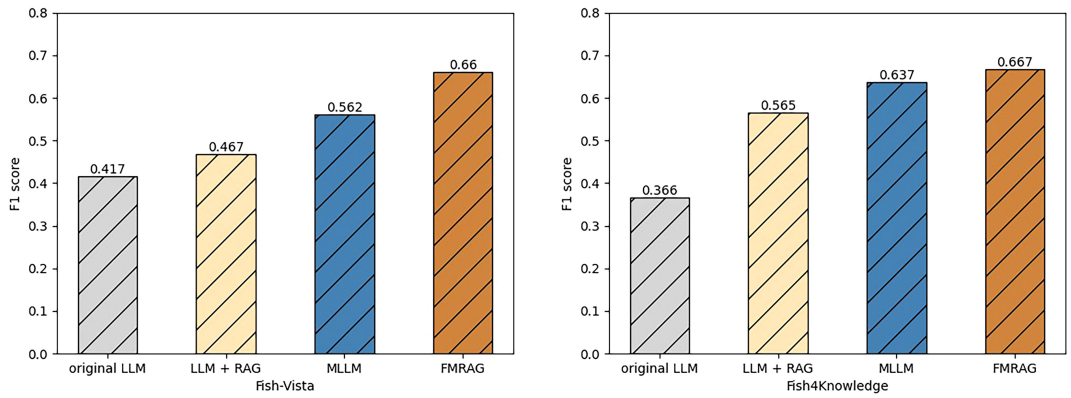


FIGURE 8
Low-frequency fish species identification performance.

TABLE 5 Comparative evaluation of fishery species detection performance between baseline mllm (trained on single images) and our fine-tuned mllm variant.

Model	FishVista		
	Precision	Recall	F1
Qwen2.5-VL-7B$_S$	0.953	0.475	0.604
Qwen2.5-VL-7B(Mfmrag)	0.914	0.867	0.852
Qwen2.5-VL-7B(Mcon+Mfmrag)	0.908	0.903	0.859
Qwen2.5-VL-7B(Mloc+Mfmrag)	0.908	0.910	0.862
Qwen2.5-VL-7B(Mloc+Mcon+Mfmrag)	0.897	0.944	0.870

TABLE 7 End-to-end hallucination detection and mitigation performance comparison.

Method	Detection acc. (%)	False positive rate (%)	Mitigation success rate (%)
Baseline MLLM (no detect)	–	–	0
Text-only Fact-Checking	72.4 ± 2.1	18.6 ± 1.8	58.3 ± 2.4
SelfCheckGPT	78.9 ± 1.7	14.2 ± 1.5	64.7 ± 2.1
FMRAG	89.2 ± 1.3	7.8 ± 1.1	84.5 ± 1.9

TABLE 6 Statistical analysis of species identification performance (mean ± std) across five random seeds.

Method	FishVista			Fish4Knowledge			WildFish		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
MLLM	0.911 ± 0.008	0.186 ± 0.012	0.361 ± 0.009	0.952 ± 0.005	0.290 ± 0.015	0.412 ± 0.011	0.951 ± 0.006	0.280 ± 0.014	0.442 ± 0.010
MLLM+RAG	0.860 ± 0.011	0.377 ± 0.018	0.515 ± 0.014	0.942 ± 0.007	0.536 ± 0.020	0.644 ± 0.016	0.941 ± 0.007	0.524 ± 0.019	0.544 ± 0.015
FMRAG	0.752 ± 0.015	0.903 ± 0.010	0.711 ± 0.012	0.950 ± 0.006	0.915 ± 0.008	0.882 ± 0.009	0.948 ± 0.006	0.914 ± 0.009	0.862 ± 0.010
FMRAG (fine-tuned)	0.781 ± 0.013	0.921 ± 0.009	0.743 ± 0.011	0.951 ± 0.005	0.959 ± 0.006	0.910 ± 0.008	0.950 ± 0.005	0.940 ± 0.007	0.890 ± 0.009

TABLE 8 Hallucination detection accuracy by error type.

Hallucination type	Text-only	SelfCheckGPT	FMRAG
Species Misidentification	68.5 ± 2.4	74.2 ± 2.1	92.1 ± 1.5
Environmental Inconsistency	71.2 ± 2.2	76.8 ± 1.9	90.4 ± 1.6
Regulatory/Policy Error	69.8 ± 2.3	75.1 ± 2.0	88.7 ± 1.7
Biomass/Quantity Overestimation	70.5 ± 2.3	77.4 ± 1.9	86.9 ± 1.8

TABLE 9 Ablation experiment performance.

Modal	Species identification accuracy	Environmental analysis accuracy	Hallucination rate
Full Modal	94.2	88.5	6.8
Text +Image	89.7	72.3	12.1
Text	76.4	65.9	18.9
Basic LLM	62.1	53.4	28.5

TABLE 10 Ablation study on impact of retrieval modalities on fmrag performance.

Retrieval image	Modality text	FishVista	Fish4Knowledge
		0.361	0.411
O		0.685	0.715
	O	0.712	0.891
O	O	0.723	0.910

TABLE 11 Enhancing fishery-mllm through multi-task fine-tuning for improved image-text association.

Fine-tuning tasks			Fish Vista	Fish4 Knowledge
Location	Concentrate	M-RAG		
			0.710	0.911
		O	0.718	0.923
	O	O	0.731	0.934
O		O	0.738	0.935
O	O	O	0.752	0.941

TABLE 12 Computational latency analysis.

Configuration	Embedding (ms)	Retrieval (ms)	Generation (ms)	Total (ms)
Baseline MLLM	–	–	850 ± 45	850 ± 45
Text-only RAG (k=5)	–	95 ± 12	892 ± 52	987 ± 58
FMRAG (k=5)	78 ± 8	112 ± 15	910 ± 48	1100 ± 62
FMRAG (k=3)	78 ± 8	68 ± 10	885 ± 44	1031 ± 51
FMRAG (k=1)	78 ± 8	25 ± 5	860 ± 42	963 ± 47

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TABLE 13 Comparison with state-of-the-art open-source MLLMs on species identification task (F1 scores).

Model	Baseline models			FMRAG (w/o fine-tune)			FMRAG (fine-tuned)		
	FishVista	Fish4 Knowledge	Wildfish	FishVista	Fish4 Knowledge	WildFish	FishVista	Fish4 Knowledge	Wildfish
LLaVA-NeXT-8B	0.352 ± 0.011	0.401 ± 0.012	0.428 ± 0.011	0.698 ± 0.013	0.871 ± 0.010	0.853 ± 0.011	0.732 ± 0.012	0.898 ± 0.008	0.881 ± 0.009
CogVLM2-19B	0.389 ± 0.010	0.434 ± 0.011	0.461 ± 0.010	0.724 ± 0.012	0.889 ± 0.009	0.868 ± 0.010	0.758 ± 0.011	0.921 ± 0.007	0.903 ± 0.008
InternVL2-8B	0.371 ± 0.011	0.422 ± 0.012	0.449 ± 0.011	0.715 ± 0.012	0.882 ± 0.010	0.861 ± 0.011	0.744 ± 0.011	0.912 ± 0.008	0.892 ± 0.009
Qwen2.5-VL-7B	0.361 ± 0.009	0.412 ± 0.011	0.442 ± 0.010	0.711 ± 0.012	0.882 ± 0.009	0.862 ± 0.010	0.743 ± 0.011	0.910 ± 0.008	0.890 ± 0.009

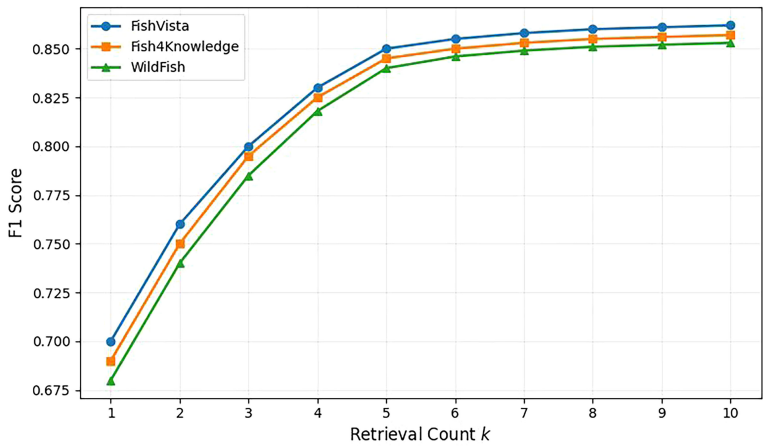


FIGURE 9
Top-k sensitivity analysis of species identification performance across three fisheries datasets.

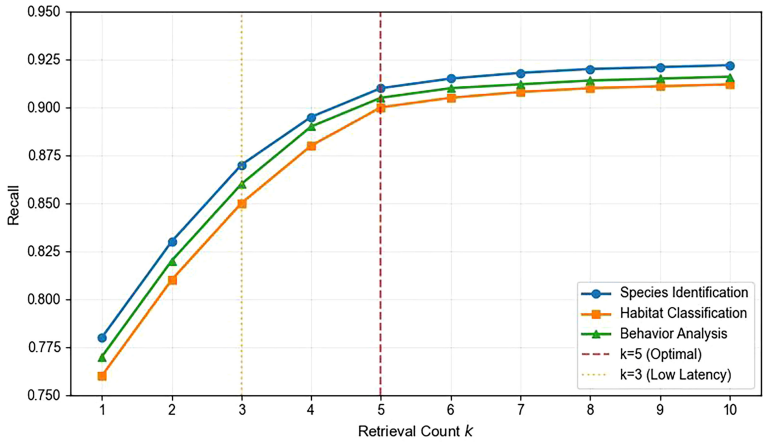


FIGURE 10
Recall@k comparison for representative fisheries tasks under different retrieval depths.

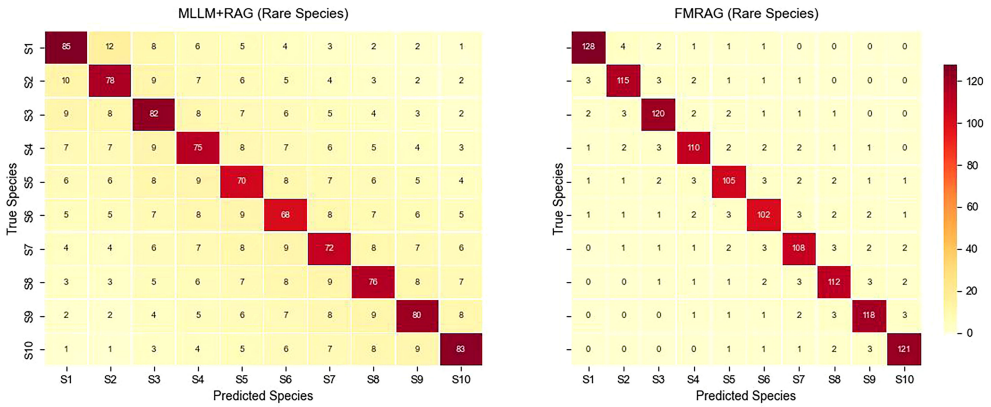


FIGURE 11
Confusion matrix comparison for the top-10 rare species: MLLM+RAG versus FMRAG.

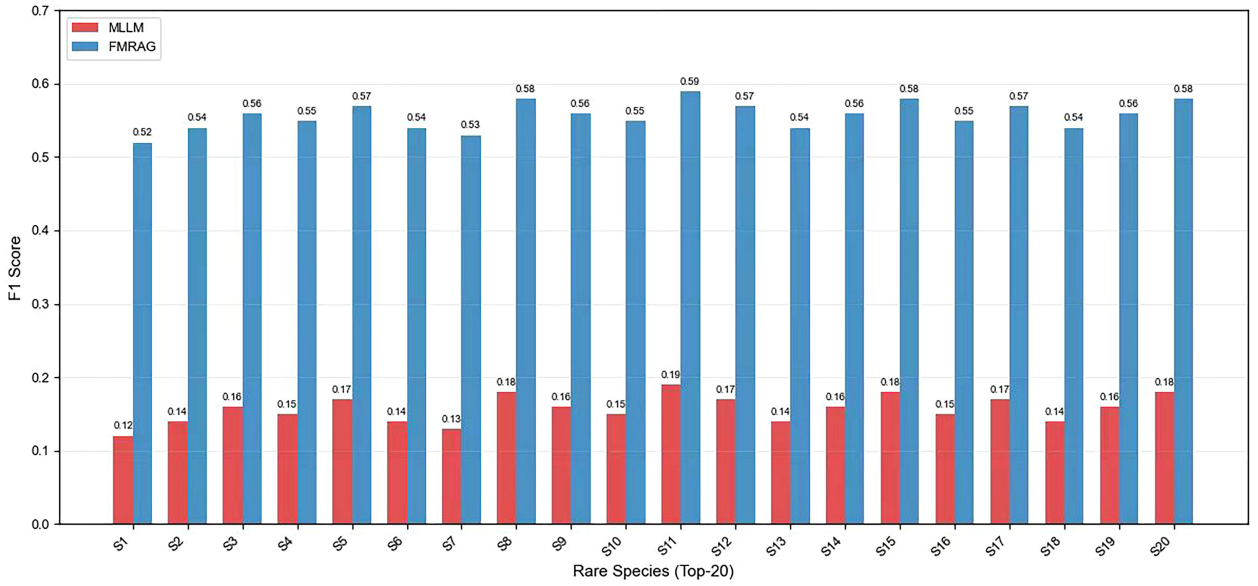


FIGURE 12
Per-species F1 comparison on the top-20 rarest species: MLLM versus FMRAG.

TABLE 14 **Rare species identification performance and statistical significance (FishVista dataset).**

Species frequency	MLLM (F1)	MLLM+RAG (F1)	FMRAG	FMRAG-FT
High-frequency (n=3121)	0.483 ± 0.012	0.634 ± 0.014	0.812 ± 0.010	0.841 ± 0.009
Rare (n=1879)	0.158 ± 0.018	0.321 ± 0.020	0.568 ± 0.016	0.598 ± 0.015
Improvement (Rare)	–	0.163	0.410	0.440

increased confidence interval coverage, and lowered temporal consistency error compared with the baseline model and text-only RAG. Figure 13 shows that biomass prediction errors became much more concentrated under FMRAG, while Figure 14 shows improved temporal stability on video sequences. Figure 15 further indicates that FMRAG provides better-calibrated confidence estimates, which is important for downstream decision support.

To rigorously evaluate whether FMRAG's improvements on rare species are statistically significant, we conducted McNemar's tests comparing classification outcomes between methods.

TABLE 15 **Comprehensive biomass estimation performance comparison.**

Method	F1 Score	MAPE (%)	ICP (%)	TCE (kg^2)
MLLM (baseline)	0.412 ± 0.015	34.2 ± 2.1	42.3 ± 2.4	128.4 ± 8.2
MLLM + Text-only RAG	0.644 ± 0.013	22.8 ± 1.8	61.5 ± 2.1	76.9 ± 5.8
FMRAG	0.882 ± 0.009	12.4 ± 1.2	84.7 ± 1.6	32.1 ± 3.4
FMRAG (fine-tuned)	0.910 ± 0.008	9.8 ± 0.9	91.2 ± 1.3	21.5 ± 2.8

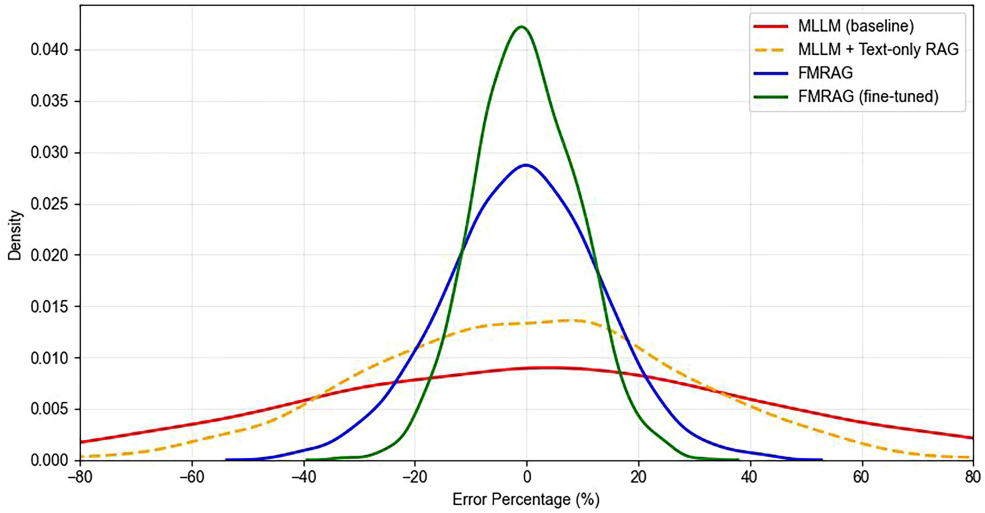


FIGURE 13
Distribution of biomass estimation errors under different retrieval-enhanced settings.

McNemar's test is particularly appropriate for paired nominal data and does not assume independence between samples. Table 14 presents rare species identification performance and statistical significance (FishVista dataset).

5 Discussion

Illegal, Unreported, and Unregulated (IUU) fishing is widely recognized as one of the most serious threats to global marine ecosystems and fisheries sustainability. According to estimates from the Food and Agriculture Organization (FAO), IUU fishing accounts for approximately 11–26 million tons of fish annually, representing a substantial proportion of global marine catches. Such activities undermine fisheries governance, distort market systems, and threaten biodiversity by enabling unsustainable exploitation of marine resources.

Detecting and monitoring IUU fishing remains challenging because relevant evidence is often fragmented across multiple modalities, including visual observations, catch reports, environmental data, and vessel activity records. In many cases,

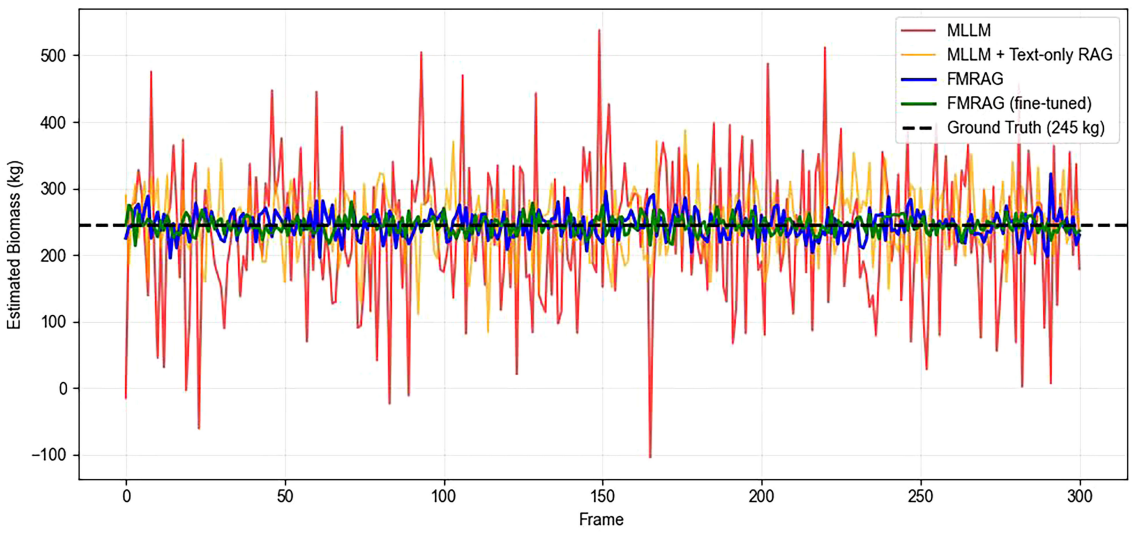


FIGURE 14
Temporal biomass estimation for a stable fish school: comparison of baseline, retrieval-augmented, and fine-tuned models.

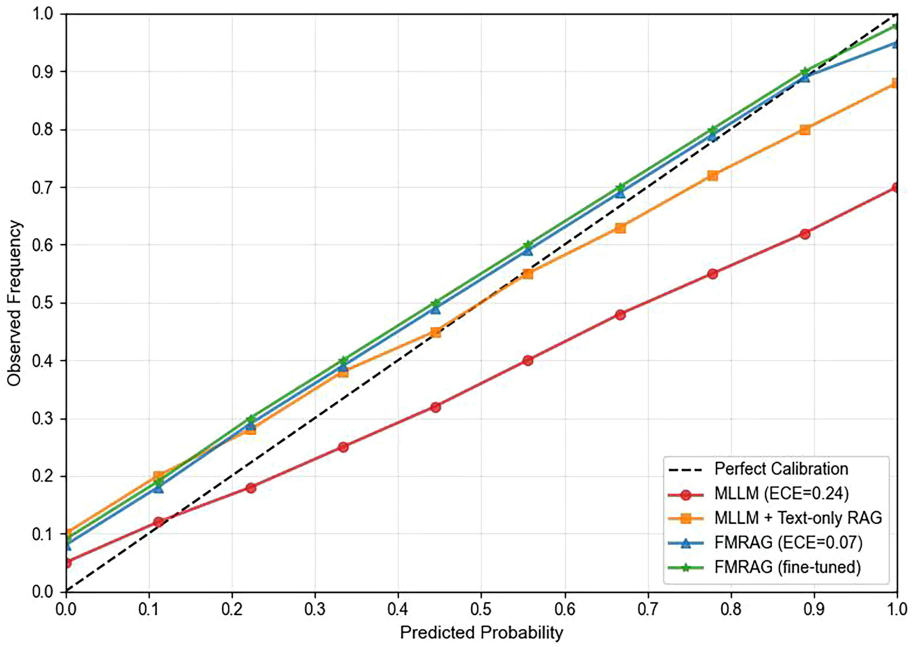


FIGURE 15
Reliability diagram of biomass confidence calibration across different model settings.

discrepancies between reported catches and observed species composition may indicate potential illegal or unreported fishing activities. Traditional monitoring systems typically rely on vessel tracking technologies such as Automatic Identification System (AIS) data or manual inspection, which may fail to capture fine-grained ecological or species-level evidence.

The multimodal reasoning capability introduced by the proposed FMRAG framework provides a potential technological pathway for improving fisheries monitoring systems. By integrating underwater imagery, textual reports, and environmental sensor data through retrieval-augmented multimodal reasoning, FMRAG can help identify inconsistencies between observed species composition and reported fisheries data. Such cross-modal verification mechanisms may contribute to detecting abnormal patterns related to potential IUU fishing activities, including misreported catches, species substitution, or anomalous ecological conditions.

6 Conclusion

This study proposed FMRAG, a fishery oriented multimodal retrieval augmented generation framework for improving the reliability of multimodal large language models in fisheries analysis. By retrieving visually similar fishery images together with their associated textual records in a shared vision language embedding space, and by integrating these retrieved references with the query input through cross modal fusion, FMRAG addresses a central limitation of existing fishery MLLMs, namely their tendency to produce hallucinated outputs under complex multimodal conditions. To further improve multi image reasoning and multimodal alignment, we introduced three fisheries specific fine tuning tasks, namely Mloc, Mcon, and Mfmrag, and incorporated a multimodal hallucination detection module for *post hoc* verification and response revision. Together, these components establish a unified framework that improves factual grounding, strengthens domain adaptation, and enhances the trustworthiness of MLLM based fisheries analysis.

Extensive experiments demonstrated that FMRAG consistently outperforms baseline MLLMs and text only RAG methods across fish species identification, rare species recognition, hallucination detection, and biomass estimation. The proposed framework achieved stronger recall and F1 performance, reduced hallucination rates, improved temporal stability and confidence calibration, and showed clear robustness across datasets, random seeds, and different open source MLLM architectures. In particular, its strong gains on rare species and noisy multimodal settings highlight its value for realistic fisheries monitoring and biodiversity related applications. Although current deployment still faces practical constraints in latency, database scale, and edge computing resources, the results show that multimodal retrieval and task specific fine tuning provide an effective path toward trustworthy fishery intelligence systems.

Future work will focus on lightweight deployment, larger scale retrieval resources, and stronger adaptability to dynamic real world environments, further extending the role of multimodal AI in sustainable fisheries management.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material. Further inquiries can be directed to the corresponding author.

Author contributions

YS: Funding acquisition, Validation, Formal analysis, Methodology, Data curation, Software, Conceptualization, Visualization, Writing – original draft. CL: Conceptualization, Investigation, Writing – review & editing, Resources, Validation, Project administration, Data curation. KZ: Writing – review & editing, Validation, Supervision, Resources, Methodology.

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Conflict of interest

Author KZ was employed by the company China Petroleum Engineering Construction Co., Ltd.

The remaining author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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